INCH-POUND
MIL-I-24768/2
25 February 1992

SUPERSEDING
MIL-P-18177C
25 May 1960

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, GLASS-CLOTH, EPOXY-RESIN (GEE)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of plies or layers of glass cloth or nonwoven parallel aligned fibers impregnated and bonded with an epoxy-resin compound or binder, oriented and processed to conform to this specification.

Intended use: The laminated material covered by this specification is intended for use in electronics applications requiring moisture resistance and high mechanical strength. The laminated material GEE is not intended for applications requiring arc and flame resistance. Caution should be used in applications involving high mechanical stresses at elevated temperatures.

Forms: Sheets, rolled tubes, and rods are recommended forms of laminated material covered by this specification.

Temperature indices: The laminated material covered by this specification has a mechanical strength test index of 140, and an electrical strength test index of 130.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/2-S-1	0.010	0.002
M24768/2-S-2	0.015	0.003
M24768/2-S-3	0.020	0.004
M24768/2-S-4	0.025	0.005
M24768/2-S-5	0.031	0.0065
M24768/2-S-6	0.047	0.0075
M24768/2-S-7	0.062	0.0075
M24768/2-S-8	0.094	0.009
M24768/2-S-9	0.125	0.012
M24768/2-S-10	0.156	0.015
M24768/2-S-11	0.188	0.019
M24768/2-S-12	0.219	0.021
M24768/2-S-13	0.250	0.022
M24768/2-S-14	0.312	0.026
M24768/2-S-15	0.375	0.030
M24768/2-S-16	0.438	0.033
M24768/2-S-17	0.500	0.036
M24768/2-S-18	0.625	0.040
M24768/2-S-19	0.750	0.043
M24768/2-S-20	0.875	0.046
M24768/2-S-21	1.000	0.049
M24768/2-S-22	1.125	0.053
M24768/2-S-23	1.250	0.055
M24768/2-S-24	1.375	0.058
M24768/2-S-25	1.500	0.061
M24768/2-S-26	1.625	0.064
M24768/2-S-27	1.750	0.067
M24768/2-S-28	1.875	0.070
M24768/2-S-29	2.000	0.073
M24768/2-S-30	2.125	0.079
M24768/2-S-31	2.500	0.085
M24768/2-S-32	2.750	0.091
M24768/2-S-33	3.000	0.097
M24768/2-S-34	3.500	0.110
M24768/2-S-35	4.000	0.122
M24768/2-S-36	4.500	0.134
M24768/2-S-37	5.000	0.142
M24768/2-S-38	5.500	0.156
M24768/2-S-39	6.000	0.172
M24768/2-S-40	6.500	0.180
M24768/2-S-41	7.000	0.195
M24768/2-S-42	7.500	0.210
M24768/2-S-43	8.000	0.216
M24768/2-S-44	8.500	0.232
M24768/2-S-45	9.000	0.240
M24768/2-S-46	9.500	0.255
M24768/2-S-47	10.000	0.270
M24768/2-S-48	—[1]	—[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/2-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Diameter of rolled round tubes: The range of sizes for rolled round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled round tube with an inside diameter of 0.125 inch and an outside diameter of 0.156 inch shall be: M24768/2-TRR-0.125/0.156. The wall thickness tolerances for rolled round tubes with an inside diameter up to 4 inches shall be as specified in table III.

TABLE II. *Range of sizes for rolled round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	48.000	0.156	50.000	0.016	1.000	1/2

TABLE III. *Wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)
0.010 to 0.015	–
0.016 to 0.030	0.006
0.031 to 0.061	0.008
0.062 to 0.124	0.009
0.125 to 0.249	0.011
0.250 to 0.500	0.013

Diameter of rods: The range of sizes and tolerances for rods shall be as specified in table IV. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.250 inch shall be: M24768/27-R-0.250.

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TABLE IV. *Size and tolerances for rods.*

Diameter (inches)	Tolerances (± inch)
0.250 to 1.999	0.005
2.000	0.008

Performance requirements: Performance requirements shall be as specified in tables V, VI, and VII.

The following requirements do not apply to this plastic laminate:

a. Tracking resistance
b. Thermal endurance
c. Silicone content
d. Flammability.

Testing: The laminated material covered by this specification shall be tested under qualification inspection requirements.

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TABLE V. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[3]	A D-48/50	4.6.4	Min. kV	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	45.0 40.0	
Dielectric constant[2]	A D-24/23 D-48/50	4.6.6	Max.	5.2 5.4	5.2 5.4	5.2 5.4	5.2 5.4 5.8	5.2 5.4	5.2 5.4	5.2 5.4	5.5 5.7	5.5 5.7	5.7
Dissipation factor	A D-24/23 D-48/50	4.6.6	Max.	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035 0.045	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035	0.025 0.035
Volume resistivity	C-96/35/90	4.6.7	Min. meghom cm				10^5						
Surface resistance	C-96/35/90	4.6.8	Min. megohm				1,000						
Arc resistance[3]	D-48/50	4.6.9	Min. seconds										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[3] Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft – lbs. per inch				7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5	7.0 5.5

See footnotes at the end of table.

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TABLE IV. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Flexural strength		4.6.12	Min. psi										
Lengthwise:	A			60,000	60,000	60,000	55,000	55,000	55,000	45,000	40,000	40,000	40,000
Crosswise:				50,000	50,000	50,000	45,000	45,000	45,000	35,000	30,000	30,000	30,000
Thermal endurance Flexural strength:		4.6.13	Min. psi										
Bonding strength	A D-48/50	4.6.15	Min. pounds							2,000 1,600	2,000 1,600	2,000 1,600	2,000 1,600
Water absorption	D_1-24/23	4.6.17	Max. percent	0.50	0.25	0.20	0.15	0.15	0.13	0.10	0.10	0.10	0.10
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]For sheets with a thickness of 0.501 inch or greater, the value for the 0.750 inch thick sheet shall apply.
[3]Maximum thickness tested shall be 2.000 inches.

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TABLE VI. *Performance requirements for rolled tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)				
					0.031–0.047	0.048–0.062	0.063–0.094	0.095–0.125	0.126–0.188
Dielectric strength	A	4.6.5	Min. volts per mil	0.125 – 8.000	350	350	250	250	200
Compressive strength[1]	A	4.6.14	Min. psi	0.125 – 0.499 0.500 – 8.000			20,000 20,000	20,000 20,000	20,000 20,000
Specific gravity	A	4.6.16	Min.	0.125 – 0.499 0.500 – 8.000	1.65 1.70	1.65 1.50	1.65 1.70	1.65 1.70	1.65 1.70
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125 – 8.000	1.0	1.0	0.8	0.7	0.6

TABLE VI. *Performance requirements for rolled tubes – Continued.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inches)				
					0.189–0.250	0.251–0.375	0.376–0.500	0.501–0.750	0.751–1.000
Dielectric strength	A	4.6.5	Min. volts per mil	0.125 – 8.000	200	150	150		
Compressive strength[1]	A	4.6.14	Min. psi	0.125 – 0.499 0.500 – 8.000	20,000 20,000	20,000 20,000	20,000 20,000	20,000 20,000	20,000 20,000
Specific gravity	A	4.6.16	Min.	0.125 – 0.499 0.500 – 8.000	1.65 1.70	1.65 1.70	1.65 1.70	1.65 1.70	1.65 1.70
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125 – 8.000	0.5	0.4	0.4	0.4	0.4

[1]Values apply to inside diameter of 0.250 inch and over and outside diameters of 2.000 inches and less.

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TABLE VII. *Performance requirements for rods.*

Requirement	Cond.	Test para.	Unit	Diameter (inches)			
				0.250–0.499	0.500–0.999	1.000	1.001–2.000
Flexural strength	A	4.6.2	Min. psi	35,000	35,000	35,000	
Compressive strength	A	4.6.14	Min. psi	35,000	35,000	35,000	35,000
Specific gravity	A	4.6.16	Min.	1.70	1.70	1.70	1.70
Water absorption	D_1-24/23	4.6.17	Max. percent	0.75	0.50	0.50	0.50

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Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
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